"When external voltage applied to the junction is in such a direction that it cancels the potential barrier, thus permitting the current flow, is called as forward biasing"

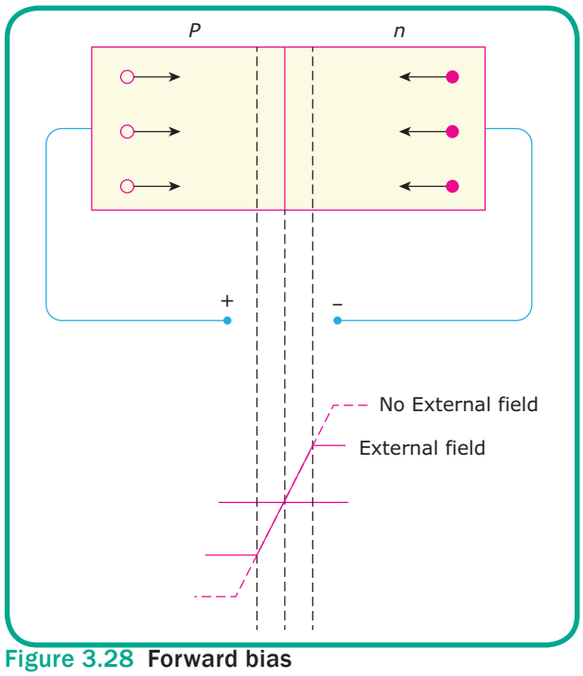


Figure 3.28 Forward bias

3.10.2 Reverse Biasing

"When the external voltage applied to the junction is in such a direction that potential barriers is increased, it is called reverse biasing"

To apply reverse bias, connect negative terminal of the battery to p-type and positive terminal to n-type as shown in Figure3.29. Because of increase in barrier potential the width of the depletion layer is also increased. As a result, the increased potential barrier prevents the flow of charge carriers (majority carriers) across the junction and hence the current does not flow.

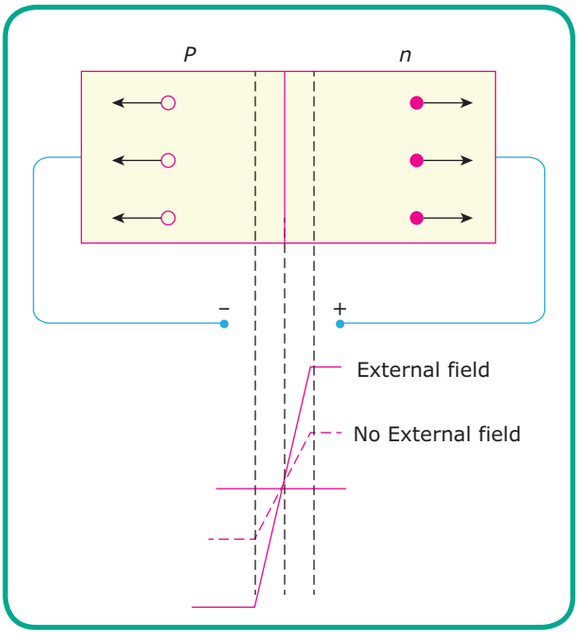


Figure 3.29 Reverse bias

From this, the students can understand the following:

1. *When pn junction is in forward biased condition, current flow occurs.*
2. *When pn junction is in reverse biased condition, no current flow occurs.*

The important terms often used with pn junction are breakdown voltage and knee voltage and are explained as follows:

T3DXNP

Breakdown voltage

"It is the minimum reverse voltage at which pn junction breakdown with sudden rise in reverse current"

Knee voltage

"It is the forward voltage at which the current through the junction starts to increase rapidly"

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